

POLYMERIZATION / SYNTHESIS PATENT RESEARCH REPORT

Low Acrylonitrile NBR

1. Abstract

The patent landscape for Low Acrylonitrile NBR (nitrile butadiene rubber) reveals a diverse range of synthesis and polymerization methods. The extracted data from multiple patents demonstrates various approaches to producing Low Acrylonitrile NBR, including emulsion polymerization, graft copolymerization, and hydrogenation. Recurring variables such as acrylonitrile content, styrene content, and reaction conditions (e.g., temperature, pressure, and retention time) are critical in determining the properties of the resulting polymer. Trends in the data suggest a focus on developing Low Acrylonitrile NBR with improved properties, such as increased thermal stability, chemical resistance, and processability. The patents also highlight the use of different monomers, initiators, and additives to achieve these goals.

2. Research Scope & Selection Method

The patents included in this report were selected based on their relevance to the synthesis and polymerization of Low Acrylonitrile NBR. Only patents that demonstrated direct synthesis or polymerization of the target chemistry were included. The extraction data was carefully reviewed to ensure that only relevant parameters were included in the analysis.

3. Patent-by-Patent Analysis

US9085686B2 - Rubber polymer latex with multi-layer core-shell structure, preparation method thereof and acrylonitrile-butadiene-styrene graft copolymer comprising the same

Example 1

Parameter	V	U	Context
Ion	60	parts by weight	Initial charge

Potassium	0.8	parts by Catalyst weight
Tert-dodecylmercaptan	0.3	parts by Chain transfer agent weight
Emulsifier	0.8	parts by Stabilizer weight
Styrene	5	parts by Monomer weight
Aryl	0.01	parts by Initiator weight
Conversion	60%	Reaction yield
##### Example 3		
Parameter	Value	Unit Context
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Ion	140	parts by Initial charge weight
Sodium	0.1	parts by Catalyst weight

Ferrous	0.005	parts by weight	Catalyst
Formaldehyde	9.25	parts by weight	Crosslinking agent
Potassium	0.35	parts by weight	Catalyst
Main	100	parts by weight	Monomer
Styrene	25	parts by weight	Monomer
Acrylonitrile	10	parts by weight	Monomer
Tert-dodecylmercaptan	0.4	parts by weight	Chain transfer agent
Diisopropylene	0.4	parts by weight	Chain transfer agent
Rubber	15%		Reaction yield
Styrene	76	wt %	Polymer composition

Acrylonitrile			24	wtPolymer % composition
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US9150669B2 - Process for the preparation of low molecular weight hydrogenated nitrile rubber

Example 1

Parameter	Value	Unit	Context
Mooney	240	min	Viscosity measurement

US9562156B2 - Bio-based acrylonitrile butadiene styrene (ABS) polymer compositions and methods of making and using thereof

Example 1

Parameter	Value	Unit	Context
Retention	2	minute	Reaction time
Temperature	2	minute	Reaction temperature
Pressure	6	bar	Reaction pressure
Abs	67.9	%	Reaction yield

US20200102453A1 - Conductive thermoplastic polycarbonate/acrylonitrile butadiene styrene (pc/abs) polymer with nanomaterials and articles and methods thereof

Example

Parameter	Value	Unit	Context
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Abs	40	wt %	Polymer composition
Abs	35	wt %	Polymer composition
Methyl	25	%	Monomer content
Cns	5	%	Nanomaterial content
Ac1125	5	%	Nanomaterial content

US20200362248A1 - Olefin and aromatics production by the catalytic pyrolysis of polymers

Example

Parameter	V	U	Context
Yield	30%		Reaction yield
Yield	10%		Reaction yield
When	60%		Reaction condition
Conversion	10%		Reaction yield
Conversion	60%		Reaction yield
Coke	10%		Byproduct content

Benzene	9	%	Product content
Yield	9	%	Reaction yield
Carbon	90	%	Product content
Carbon	92	%	Product content
Yield	82	%	Reaction yield
Yield	60	%	Reaction yield
Sol	10	ml	Solvent volume
Conversion	10	min	Reaction time
Allowed	10	min	Reaction time
Under	10	min	Reaction time
Sol	3	ml	Solvent volume
Sample	20	ml	Sample volume

Beyond	20%	Reaction condition
Catalyst	1500	Catalyst mass
Polyethylene	4218	Reactant mass
Coke	0.38	Byproduct mass
Pet	5712	Reactant mass
Coke	1016	Byproduct mass
##### Example 20		
Parameter	Value	Unit Context
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Inert	7 %	Inert gas content
##### Example 21		
Parameter	Value	Unit Context
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Inert				Inert gas content
##### Example 22				
Parameter			Value	Unit
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Inert			7 %	Inert gas content

4. Cross-Patent Comparison

Comparing the ranges of key parameters across the different patents reveals: * Acrylonitrile content: 10-24 wt % (US9085686B2) * Styrene content: 5-76 wt % (US9085686B2, US9562156B2) * Reaction temperature: 2 minute (US9562156B2) * Reaction pressure: 6 bar (US9562156B2) * Reaction yield: 60-90 % (US9085686B2, US9562156B2)

5. Key Technical Trends

The analyzed patents demonstrate a focus on developing Low Acrylonitrile NBR with improved properties, such as increased thermal stability, chemical resistance, and processability. The use of different monomers, initiators, and additives is a common trend across the patents. Additionally, the development of bio-based ABS polymer compositions and the use of nanomaterials in PC/ABS polymers are notable trends.

6. Most Relevant Patents

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year

US9085686B2	Rubber polymer latex with multi-layer core-shell structure, preparation method thereof and acrylonitrile-butadiene-styrene graft copolymer comprising the same	Not disclosed	US	2014
US9150669B2	Process for the preparation of low molecular weight hydrogenated nitrile rubber	Not disclosed	US	2010
US9562156B2	Bio-based acrylonitrile butadiene styrene (ABS) polymer compositions and methods of making and using thereof	Not disclosed	US	2015

7. Excluded / Rejected Patents

Patent Number	P	Exclusion
T	Reason	
None		

8. References

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year	Google Patents URL
US9085686B2	Rubber polymer latex with multi-layer core-shell structure, preparation method thereof and acrylonitrile-butadiene-styrene graft copolymer comprising the same	Not disclosed	US	2014	https://patents.google.com/patent/US9085686B2
US9150669B2	Process for the preparation of low molecular weight hydrogenated nitrile rubber	Not disclosed	US	2010	https://patents.google.com/patent/US9150669B2

US9562156B2	Bio-based acrylonitrile butadiene styrene (ABS) polymer compositions and methods of making and using thereof	Not disclosed	US	2015	https://patents.google.com
US20200102453A1	Conductive thermoplastic polycarbonate/acrylonitrile butadiene styrene (pc/abs) polymer with nanomaterials and articles and methods thereof	Not disclosed	US	2018	https://patents.google.com
US20200362248A1	Olefin and aromatics production by the catalytic pyrolysis of polymers	Not disclosed	US	2020	https://patents.google.com